

7.2.7 Software Audit Process

NOTE — The Software Audit Process provides **independent determination** of the degree to which software products and processes comply with specified requirements, plans, and agreements. It supports objective assessments, identifies non-conformances, and ensures follow-up actions are taken and verified.

7.2.7.1 Purpose

The purpose of the Software Audit Process is to **independently determine compliance of selected products and processes with the applicable requirements, plans, and agreements**, as appropriate.

7.2.7.2 Outcomes

As a result of successful implementation of the Software Audit Process:

- a) an audit strategy is developed and implemented;
 - b) compliance of selected software work products and/or services or processes with requirements, plans, and agreements is determined according to the audit strategy;
 - c) audits are conducted by an appropriate independent party; and
 - d) problems detected during an audit are identified and communicated to those responsible for corrective action and resolution.
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7.2.7.3 Activities and Tasks

The project shall implement the following activities in accordance with applicable organizational policies and procedures with respect to the Software Audit Process.

7.2.7.3.1 Process Implementation

This activity consists of the following tasks:

7.2.7.3.1.1 Audit Scheduling

Audits shall be scheduled at **predetermined milestones**, as specified in project or quality assurance plans. Additional audits may be conducted when significant changes occur or at the request of stakeholders.

7.2.7.3.1.2 Auditor Independence

Auditing personnel shall not have any **direct responsibility** for the software products and activities they audit. They shall have sufficient authority to access necessary information and raise findings without conflicts of interest.

7.2.7.3.1.3 Audit Resources

All resources required to conduct the audits shall be agreed upon by the parties. These resources may include support personnel, locations, facilities, hardware, software, tools, and access to documentation and repositories.

7.2.7.3.1.4 Audit Preparation and Agreement

Prior to each audit, the parties shall agree on the following:

- agenda;
 - software products and/or process outputs to be reviewed;
 - audit scope and procedures;
 - applicable standards and checklists;
 - **entry and exit criteria** for the audit.
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7.2.7.3.1.5 Problem Recording and Resolution

Problems detected during the audits shall be recorded and entered into the **Software Problem Resolution Process (7.2.8)** for tracking and resolution.

7.2.7.3.1.6 Audit Reporting

After completing an audit, the audit results shall be **documented and provided to the audited party**. The audited party shall acknowledge to the auditing party any problems found and describe planned resolutions.

7.2.7.3.1.7 Audit Closure

The parties shall agree on the **outcomes of the audit**, responsibilities for action items, and **closure criteria** for findings. Closure of audit findings shall be verified and documented.

7.2.7.3.2 Software Audit

This activity consists of the following task:

7.2.7.3.2.1

Software audits shall be conducted to ensure that:

- a) as coded, software products (such as software items) reflect the approved design documentation;
 - b) acceptance review and testing requirements prescribed by the documentation are adequate for software product acceptance;
 - c) test data comply with specifications;
 - d) software products were successfully tested and meet their specifications;
 - e) test reports are correct and discrepancies between actual and expected results have been resolved;
 - f) user documentation complies with applicable standards;
 - g) activities have been conducted according to applicable requirements, plans, and contracts;
 - h) costs and schedules adhere to the established plans.
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7.2.7.4 Deliverables

The following deliverables shall be produced during this process:

- Software Audit Strategy or Plan (if separate);
 - Audit schedules and agendas;
 - Audit checklists and criteria;
 - Audit reports and findings logs;
 - Records of non-conformances entered into the Problem Resolution system;
 - Audit closure records and evidence of corrective actions;
 - Independent auditor statements or certifications (if applicable).
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7.2.7.5 Audit and Configuration Control

All audit plans, reports, logs, and findings shall be placed under **configuration management**.

Audit evidence shall be retained in accordance with organizational and regulatory requirements.

Digital signatures or equivalent mechanisms shall be applied to preserve authenticity and integrity.

Audit records shall be accessible to authorized parties for traceability and compliance audits.

Cross-References

This process interacts with the following:

- **Software Requirements Analysis (7.1.2)** — provides baseline requirements for audit checks.
- **Software Construction (7.1.5), Integration (7.1.6), and Qualification Testing (7.1.7)** — provide work products and test evidence subject to audit.
- **Software Quality Assurance (7.2.3)** — coordinates audit scheduling and scope as part of QA planning.
- **Software Verification (7.2.4)** and **Validation (7.2.5)** — provide artifacts and results that audits examine for compliance.
- **Software Review (7.2.6)** — complements audits with peer evaluations during development.
- **Software Problem Resolution (7.2.8)** — receives problems and non-conformances found during audits for resolution tracking.